

A national survey of persistent, bioaccumulative, and toxic (PBT) pollutants in the United States milk supply.



This study measured 21 persistent, bioaccumulative, and toxic (PBT) pollutants in the US milk supply. Since milk fat is likely to be among the highest dietary sources of exposure to PBTs, it is important to understand their levels in this food. Nationwide samples were collected from 45 dairy plants in July of 2000 and again in January 2001. The levels of all chemicals in the chlorobenzene, pesticide and other halogenated organic groups were determined to be below their detection limits in all samples. National averages were computed for 11 chemicals or chemical groups found above the detection limits. The national average CDD/CDF and PCB TEQ concentrations were 14.30 and 8.64 pg/l, respectively, for a total of 22.94 pg/l. These levels are about half the values found in a similar study conducted in 1996. If this difference is in fact indicative of declining milk levels and assuming exposure levels from nondairy pathways have remained the same over this time period, this would result in an overall decrease in adult background dioxin exposure of 14%. Six PAHs were detected with national averages ranging from 40 to 777 ng/l. Cadmium concentrations ranged from 150 to 870 ng/l with a national average of 360 ng/l. Lead concentrations were consistently higher than those of cadmium, ranging from 630 to 1950 ng/l with a national average of 830 ng/l. PAHs showed the strongest seasonal/geographic differences, with higher levels in winter than summer, north than south and east than west. Average adult daily intakes from total milk fat ingestion were computed for all detected compounds and compared to total intakes from all pathways: CDD/CDF/PCB TEQs: 8 vs. 55 pg/day, PAHs: 0.6 vs. 3 micro g/day, lead: 0.14 vs. 4-6 micro g/day, and cadmium: 0.06 vs. 30 micro g/day.